

Synopsis: A Qualitative Cosmology under Hysterical Response Theory

The aeon loop and the galactic subplot, without the narrative padding

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Computational note. Exploratory experiments and paper writing were assisted by generative AI. This synopsis inherits claims from companion technical papers; it states no new formal theorems for Lean verification.

Abstract

A one-sitting map of the qualitative cosmology paper: the closed Hysterical aeon, exact conformal boundary matching, and the galactic formation chain $Q \gtrsim O(\sqrt{M_b}) \rightarrow \text{stable attractor} \rightarrow \text{spherical-to-oblate halo} \rightarrow R \sim M_b^{5/12}$.

1 Why a qualitative cosmology

The technical companions prove local mechanisms. This synopsis (and its full qualitative companion) only arranges them into one chronological universe.

2 Ontology

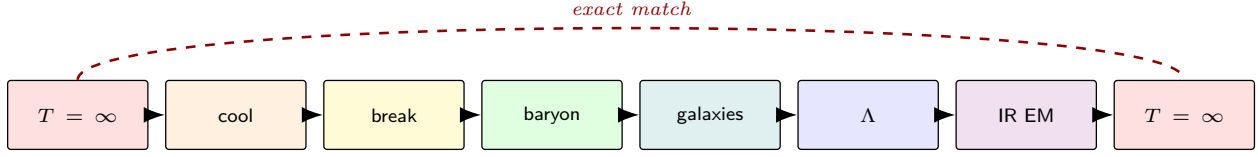
$$\mathbf{F}_{\text{observed}} = \mathbf{F}_{\text{ordinary}} + \mathbf{F}_H, \quad \mathbf{F}_H = -\text{Tr}(F\mathcal{L}^{-1}\mathcal{S}).$$

No new force carrier: residuals of gravity, electroweak dynamics, and electromagnetism, each critical in its epoch.

3 The aeon cycle

$$\begin{aligned} T = \infty \text{ (hot EM)} &\rightarrow \text{cooling} \rightarrow \text{symmetry breaking} \rightarrow \text{baryogenesis} \\ &\rightarrow \text{early galaxies} \rightarrow \Lambda\text{-like attractor} \rightarrow \text{IR EM hysteria} \rightarrow T = \infty. \end{aligned}$$

Under reciprocal conformal matching, the outgoing electromagnetic future and the incoming hot beginning are the *same* surface: the cycle boundaries match exactly.



4 Phase anchors (order of magnitude)

- Hot EM beginning: incoming radiation branch of HCCC conformal matching.
- EW concurrency: $t_\diamond \sim 10^{-11}$ s when $T_\diamond > T_*$.
- Early galaxies: nonlinear contrast by $z \simeq 14$ possible for strong gravitational response coupling (timing, not abundances).
- Late Λ -like term: dilution-driven criticality and vacuum-like attractor.
- End of aeon: infrared-first EM Hysterical instability $\rightarrow$ active crossover.

5 Galactic subplot

1. **Critical gravity.** Positive matter–response coupling $\Rightarrow$ supercritical frozen Jacobian; growth above ordinary dust.
2. **Capacity.** Formation/retention conjecture: $Q \gtrsim O(\sqrt{M_b})$.
3. **Relaxation / stability.** Capacity–filling ODE $\rightarrow$ stable attractor $Q^2 \propto M_b$ (hence BTFR $v_f^4 \propto M_b$ with flat $v_f^2 = CQ$).
4. **Spherical $\rightarrow$ oblate.** Leading halo $n \propto Q/r^2$ is spherical; first angular correction is an oblate quadrupole.
5. **Size–mass.** Isotropic accretion $\Rightarrow R \propto M_b^{5/12}$ ($\alpha = 5/12$).

6 What is not claimed

Observed η_B magnitude; galaxy abundances; saturated IR EM amplitude; junction regularity; proof that Nature realises the whole cycle. Those remain open or outside the inherited theorems.

7 Pointer

Full qualitative narrative, figures, and status table: `paper_14_hysterical_universe.tex`.